# Rishika Mudundi

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## Education

**Texas A&M University** **Aug 2024–May 2028**
*Bachelor of Science in Computer Engineering (GPA: 3.7/4.0)* *College Station, TX*
**Coursework:** Data Structures & Algorithms, Computer Architecture (Belfast, UK), Digital Design Systems, Circuit Theory

## Awards

ECE Junior Research Scholar | Dean's Honor Roll (Fall '25) | Paul '92 & Michelle Deere '92 EE Scholarship

## Experience

**Junior Research Scholars Program (JRS)** **Aug 2026–Present**
*Texas A&M University SETH Lab* | *Undergraduate Researcher* *College Station, TX*

- Conduct post-silicon validation and user-mode differential testing on RISC-V CPUs, adapting the RISCover fuzzing framework and testing architecture to systematically target edge-case security vulnerabilities
- Construct a 4-board RISC-V hardware testbed with an FPGA host controller, handling Linux flashing, static IP network configuration, UART troubleshooting, and SD card storage integration

## Projects

**C++ Cache Architectural Simulator** **Jul 2026–Aug 2026**

- Developed a configurable C++ cache simulator modeling direct-mapped, set-associative, and fully associative architectures with LRU cache line eviction and miss penalty cycle counters
- Benchmarked cache hit/miss counts and execution performance across 10 simulation runs, demonstrating that array-based indexing consistently outperformed hash maps in execution time
- Implemented Object-Oriented design principles and managed command-line development workflows via Git

**FSM Traffic Light Controller** **Dec 2025**

- Implemented and synthesized a Mealy Finite-State Machine (FSM) in Verilog on a Xilinx FPGA, mapping board buttons as dynamic sensor inputs and onboard LEDs as physical light states for a traffic intersection
- Expanded Verilog logic by integrating an additional sensor input into the state conditions to alter light sequence

**Embedded Posture Detector** | **TAMUhack 2026** **Jan 2026**

- Programmed an Arduino to sample real-time ultrasonic sensor data, implementing threshold-based triggers and audio buzzer alerts when poor posture persisted beyond 3 seconds
- Integrated EEPROM non-volatile memory to capture and retain posture alerts for an HTML/CSS progress tracker

## Activities

**MSC Visual Arts Committee** **Aug 2024–Present**
*Texas A&M University* | *Gallery Director* *College Station, TX*

- Led cross-campus outreach and gallery space logistics to curate artwork for campus exhibits, organizing student volunteers and artist timelines across 5-6 week planning cycles

**Residence Life Utilities Challenge** **Jan 2026–Apr 2026**
*Texas A&M University* | *1st Place* *College Station, TX*

- Awarded **1st Place** in a 3-person team competition by analyzing multi-year utility consumption data for a 26k+ sq. ft. community center, proposing a digital brochure system with projected savings of $5,000+ annually

**Society of Women Engineers** **Aug 2024–Present**
*Texas A&M University* | *Member* *College Station, TX*

- Organized STEM conference activities and hands-on engineering challenges for 100 high school students

## Technical Skills

**Hardware**: Verilog, FPGA (Xilinx), Digital Logic
**Languages & Toolchains**: C++, Python, Linux, Git, ARMv8 Assembly, GDB, HTML, CSS
**Lab & EDA**: Oscilloscope, Multimeter, DC Power Supply, Multisim, CAD